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Investigation on Preserving Privacy of Electronic Medical Record using Split Learning

Kiruthika M[a], Abhishek Kumar[b], Lalitha Krishnasamy[c], Velliangiri Sarveshwaran[a]

[a]*Department of Computational Intelligence, School of Computing, SRM Institute of Science and Technology, Kattankulathur, Tamilnadu 603203, India*

[b]*Senior Manager IT, Punjab National Bank, New Delhi - 110075, India*

[c]*Department of Computer Science and Engineering, School of Engineering and Technology, Christ University, Bengaluru, Karnataka 560074, India*

velliangiris@gmail.com

Abstract

Artificial Intelligence is deployed in multiple areas, including healthcare. Utmost research is done in AI enabled healthcare industry because of the demands like accurate result, data security, exact prediction, huge volume of data, etc. In conventional deep learning models, the training happens with the dataset that are stored in a single device. This requires a huge storage space and highly efficient machines to train the data. Usage of big data, demands for innovative models that can be deployed and used in confined storage. Split learning is one such collaborative distributed deep learning model that allows the data to be stored in a split fashion. Split learning supports desirable features like less storage, more privacy to raw data, ability to work with resource constraints, etc., making it suitable for storing electronic medical record of patients. This paper discusses the advantages of using split learning for healthcare, the possible configurations of split learning that supports data privacy in healthcare and finally discusses the open research challenges in implementing split learning for healthcare.

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Keywords: Split Learning; Healthcare; Privacy; Security; Patient data;

1. Introduction

Being in the era of Information Age, Artificial Intelligence (AI) plays a vital role in everyone's life by simulating Human Intelligence. AI is applied in major domains like Finance, Marketing, Social media, Healthcare, Manufacturing Industries and much more. Of all these, major developments are carried over in healthcare industry using AI.

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As forecasted by Statista [1] platform, the world-wide AI based healthcare market would reach nearly 188 billion U.S. dollars by 2030. Such an enormous development in AI Healthcare depends completely on data, as working of AI starts with training huge amount of data, through which it learns and gains intelligence. With this intelligence, AI analyzes the real-world data and benefits Healthcare industry in every possible way. In healthcare industry, AI is applied for various functionalities like Image based Diagnosis, Patient Monitoring, Robotic Surgeries, Administrative operations, Supply Chain Management, Disease Prediction, etc., as in Fig 1.

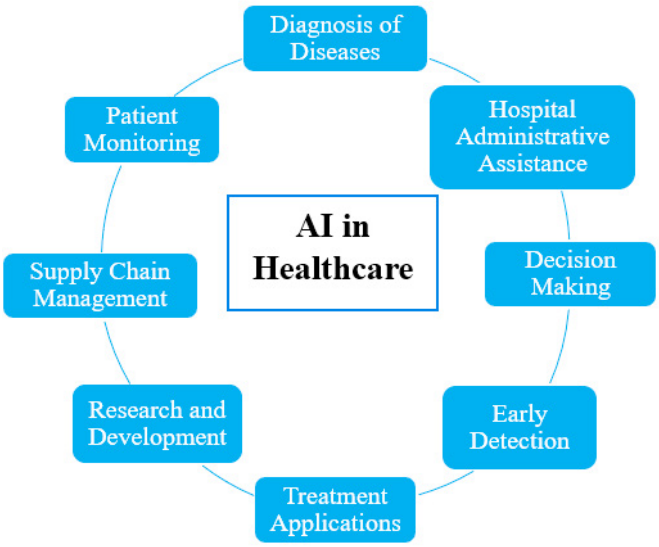


Fig. 1. Role of AI in Healthcare

It is more familiar that healthcare industry handles huge volume of patient's medical record, that may include demographic related information, prevailing symptoms, complete history of diagnoses, disease comorbidities, prescribed medications, surgeries / procedures undergone, radiographs, laboratory reports, etc. In addition to clinical data, staff registries, supply chain information, equipment / machinery information, administrative details, payroll data, R&D oriented data and even more data are also handled. The more the data is used, the more is the threat to the data. Especially, healthcare data is highly important for several reasons like – Sensitivity of Patient's data, Compliance to healthcare policies, Financial losses due to cyber-attacks, and so on [2]. Patient's Electronic Medical Record (EMR) has become more vulnerable to data breaches and cyber-attacks that can result in substantial financial loss and lack of trust [3]. Hence, in healthcare industry, it is most important to preserve the patient's medical record. There are multiple ways to protect EMR. Few techniques include Encryption, Access Control & Authorization, Data Obfuscation / Masking / Tokenization, Catastrophic Management / Recovery, etc. AI needs to handle large volumes of data including medical images and videos. Few problems that arise when handling large volumes of data involves data security, data storage and retrieval issues. More focus is given for developing novel algorithms for AI model and least research is carried out in the sector of data privacy in AI.

Multiple attacks are induced on an AI dataset including Poisoning / Tampering attack, Adversarial attack, Inference attack, Watermark attack, Denial of Service attack, Zero-day attack, Cyber Physical attack etc., [4 – 6]. Split learning (SL) can be considered as a solution to the problems related with data handling as data is not accumulated in a central repository in SL. In SL, AI learning does not happen with consolidated dataset. The concept of Split learning was developed at MIT Media Lab that enables the AI entities to train the machine learning model with locally available data, as in Fig 2. In other words, Split learning is a collaborative machine learning approach where ML models are split in training but collaborates with each other to give the final result [7]. The data resides in the local system itself. Split learning works based on Client - Server model. Deep learning models [8] like Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Recursive Neural Network (RecNN), Deep Belief Network (DBN), Autoencoders, Radial Basis Function Network (RBFN), etc., works on multiple layers. The attractive part of Split

learning is that it works by partitioning these deep learning model layers in a such a way that the work of some layers is carried over in the client end while the work of the rest of the layers are done in the server end. Hence, no raw data is shared over the network and the models are trained in a distributed manner. Also, the clients are not loaded with too much computation and complexity.

SplitNN is the model used for training the neural network in a split fashion across the network. The split model is self-contained in the sense that each model trains the data until the cut layer (last layer in client side) and produces an intermediate result to its connected model. The weights are transferred from the local machine to the central machine and no raw data is shared and the training is carried over in the central machine Though the entities involved in Split learning does not know about each other still the entities train their model with the data in hand.

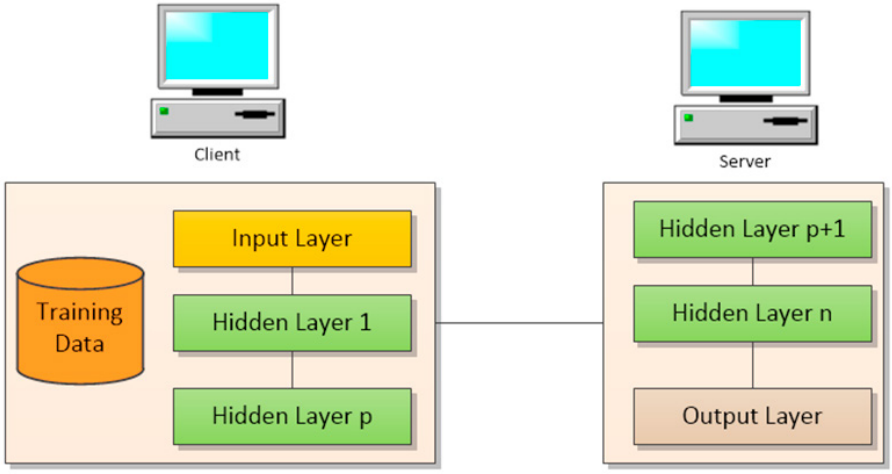


Fig. 2. Split Learning Model

After receiving the partial / intermediate trained models from the client, the servers complete the training and calculated the gradients. These gradients are backpropagated to the clients for the next training iteration. This process repeats until the global model is completely trained. Split learning model can be successfully applied in any sectors including healthcare, manufacturing, logistics, governance and finance. Split learning can be implemented in healthcare where local / small hospitals with few datasets can build a machine learning model with their own data and this model can be collaborated with other models from various other hospitals in order to build a superior model. It is ultimately important to note that no raw data is shared in the entire process of model building and no privacy regulation is overruled.

The work is organized as follows: In the next part, the related works carried out by other researchers in the area of split learning are discussed with the benefits and drawbacks. The next section consolidates the various split configurations that can be used for sharing information between client and server. Finally, the current research challenges in developing split learning are identified and that eventually acts as a roadmap for further researches in this area.

2. Related Work

This section provides an insight about the split learning model and its use cases specifically in healthcare industry. Few SplitNN configurations like Simple vanilla split learning, Split learning without label sharing and Split learning for vertically partitioned data are proposed in [9] to facilitate distributed learning. These configurations are implemented in healthcare dataset and the during the process of learning, labels and raw data are not shared. The performance of SplitNN configurations is compared with other distributed learning methods like Federated Learning (FL) and Large batch Synchronous stochastic Gradient Descent (LS-SGD) learning. The comparisons show that SplitNN consumes the resources efficiently and performs way better than other distributed learning architectures.

Though, the novel configurations were introduced, the research fails to discuss about the limitations and not much details regarding implementation are given. In [10], split learning was presented as a novel mechanism for training deep neural networks without actually revealing the raw data to the communicating nodes. The work focusses on semi-supervised training models and the proposed algorithm use computer vision datasets like MNIST, CIFAR-10 and ILSVRC 2012. The algorithm clearly states the steps in which the distributed neural network works with minimal resources. This method is highly useful in places where there is no legal permission to share the original data as raw data is not shared. Each dataset is trained in different physical localities and are combined later to give the classification results.

Split learning is applied to medical data in [11]. U-shaped configuration was used here and data from three different hospitals that are in distributed locations are allowed to train a collaborative model without sharing raw data. The performance of split learning is compared with both centrally hosted data and non-collaborated data. The dataset contains images and multi label dataset of chest x-rays. The results clearly shows that split learning supported scalability and was performing better even when the number of clients increase over time. This research failed to address the details about the configuration used for performing split learning. Also, the results completely depend on the dataset considered.

A spatio-temporal machine learning model was presented in [12] where the model considers spatially distributed clients and a central server. A privacy preserving layer is deployed in the client side while hidden and output layers are deployed in the server side. The main datasets considered are COVID-19 chest scans and Musculoskeletal Radiographs (MURA) and cholesterol levels. The clients send the encrypted feature maps to the server. The method claims to be versatile and so it can be deployed in any other sectors like banking, finance, insurance, etc. The main observation is that the model fails to address the encryption method used for encrypting the feature maps, that is shared from the client to the server. Also, there is no specification as how the backward propagation happens.

Based on IoT devices, a comparison study between federated learning (FL), split learning (SL), and split-fed learning (SFL) was conducted in [13]. The used datasets are divided into imbalanced, identical, and non-identical data. Since IoT devices are being deployed, greater emphasis is placed on accuracy and convergence speed while taking resource constraints into consideration. The end results demonstrate that SFL requires less training time than SL and less computational overhead than FL. But SFL's performance is still controversial given that IoT devices have additional resource limitations, with respect to memory.

Vertical Split Learning (VSL) concept is introduced in [14] that partitions the dataset vertically and learning happens based on distributed architecture. This model proposes to work better in both medical and non-medical datasets. In most cases hospital databases are maintained in a distributed basis and this method makes use of feature splitting. VSL is defined as a better alternative to the centralized learning but occasionally VSL may underperform because of the communication issues, non-relevancy in partial data being processed, lack of efficiency in learning, no exact match between the vertical data, etc.

Internet of Things based Split Learning (IoTSL) algorithm is developed in [15]. This framework works in a collaborated manner for IoT devices by employing Generative Adversarial Network (GANs) to train the data locally. Initially GANs work with available global data and model is fine-tuned with local training. GAN usually works with two sub-models namely Generator and Discriminator. Generator generates new examples and Discriminator classifies the output of the generator to be real or manipulated. The model claims to incur less communication cost and produce accurate results. Since this model is deployed for IoT devices, using GAN may not be a good choice as generators contribute to additional processing cost and even when it is deployed in a distributed manner, still the IoT devices must be trusted.

To safeguard shared information in crucial networked infrastructures like wireless networks, an advanced intrusion detection system based on machine learning is introduced in [16] that makes use of Split learning. The model offers improved accuracy, improved performance in detection and classification. Split Learning-based IDS is proposed for Intelligent Transportation System (ITS) and is compared with models that practice Federated Learning and Transfer Learning. Various parameters are considered for comparison such as Accuracy, Detection rate, ROC Curve, Packet Delivery Ratio, Power Consumption rate, Quality of Experience and Delay Analysis. A client – server approach is used for transferring the weights and gradients. A novel technique known as Masked Split Learning (MaskSL) is developed in [17]. This method enhances security and pre-trains the sensitive images based on mask. The model is

trained by dividing it into two parts – encoder and decoder. Encoder has embedding layer and transformer layers while decoder has transformer layers and Multi-Layer Perceptron layer. The image masking mechanism uses Fisher information and Differential Privacy for masking and securing. . Comparison study of Split leaning model with respect to healthcare industry is shown in Table 1.

Table 1. Comparison study of Split leaning model with respect to healthcare industry.

References	Key Contribution(s)	Observation(s)
[9]	SplitNN Configurations - Simple vanilla split learning, Split learning without label sharing, Split learning for vertically partitioned data	Lacks detailed explanation of the configurations with respect to patient data
[10]	Distributed neural network training architecture is used to train the physically split dataset.	Training of data is done without sharing the labels. Almost U-shaped gradient sharing method is used.
[11]	Split learning using U-shaped configuration is used for sharing ant training hospital data.	No details about how the training happens and comparison is not done with recent developments.
[12]	Spatially distributed clients interact with a server and perform split learning while preserving privacy in the client end.	There are no details about the encryption mechanism used for encrypting feature maps.
[13]	New learning technique called SplitFed Learning is proposed for IoT devices.	It is still debatable if SFL works better because of the resource constraints that prevails in IoT devices.
[14]	Dataset is vertically split so few features / attributes of a patient lies in one machine and other features lies in one or more machines.	VSL may underperform based on the dataset as few data may not match perfectly with other available data.
[15]	IoTSL uses GAN algorithm in a distributed way to train the dataset in IoT devices.	Generators used in GAN may require additional computation cost.
[16]	Split learning-based IDS is designed for detecting the attacks in Intelligent Transportation System.	The performance of the algorithm may degrade for heterogenous networks. Also the architectural design of the transfer of weights is not clearly explained.
[17]	Highly sensitive images are pre-trained using MaskSL that has Encoder & Decoder layers. IT makes use of Fisher information and Differential Privacy to provide privacy of data / images.	The algorithm works better for large scale devices. Further optimization is required for reducing the communication and computational overheads.

3. Split Learning Configurations for Data Privacy

Few configurations of SplitNN are discussed below so as to understand the working of split learning environment and all the different ways for implementing the split learning [18]. The first and foremost is the Simple Vanilla Split learning configuration, where the clients train the raw data until a cut layer and then the smashed data and labels are sent to the server end, as in Fig 3. The server carries out the remaining part of training and processes until output layer. The error is calculated and the gradients are propagated back to the client, where the process is repeated. Benefit: Simple, Drawback: Shares label.

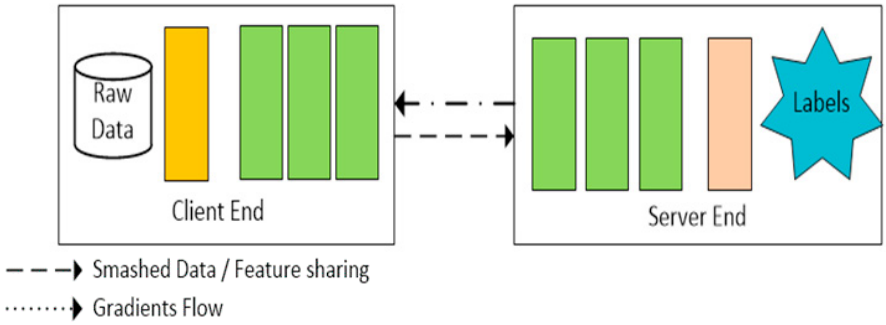


Fig. 3. Simple Vannila Split Learning

In Vertical Split learning configuration, each client has a partition of a dataset that is split based on some features and it is represented in Fig 4. Each client will process its partitioned data until the cut layer and forward propagates the smashed data to the server. The server receives the smashed data from all the clients and concatenates it and process the data until output layer. Once the error is calculated, the server back propagates the gradients to the respective clients. In this configuration also, label is shared with the server and raw data is not shared.

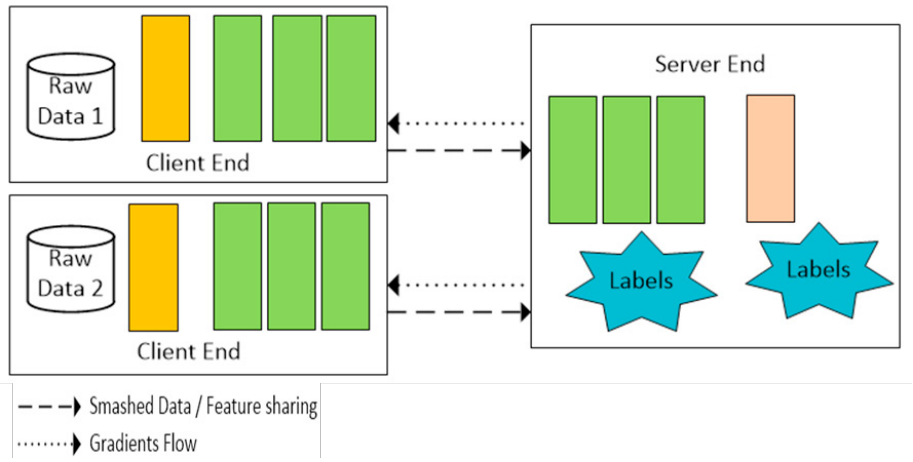


Fig. 4. Vertical Split Learning

U-shaped Split Learning model is the next SL configuration considered, and in this model, the client neither exchange raw data nor labels with the server, as in Fig 5. The data and labels are maintained by the client itself. In this case, one client holds the data while other clients and servers train the smashed data that the client sends. The forward propagation continues until the server-cut layer, at which point it is activated and sent to the client. The client is now finished with forward propagation up to the output layer. The gradients are then sent to the server and backpropagation is started. Now the server delivers the shattered data back to the client while carrying out the backpropagation. This way, the forward and backward propagation are executed in a U-shape fashion.

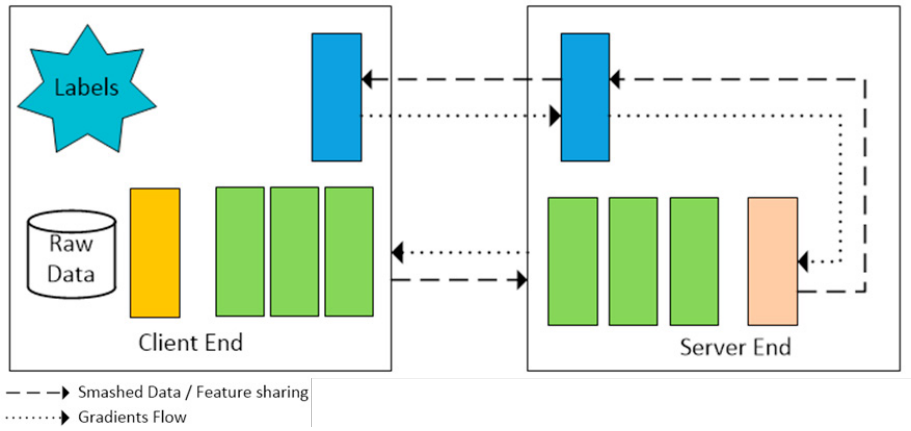


Fig. 5. U-Shaped Split Learning

Another SL model is Extended Vanilla Split Learning where additional clients are used for processing the intermediate layers in the set of hidden layers. This additional processing is done before passing the smashed data to the server. Here, it is assumed that only the smashed data is passed on between coordinating clients and the server and the data actually resides with one client only. This model is shown in Fig 6.

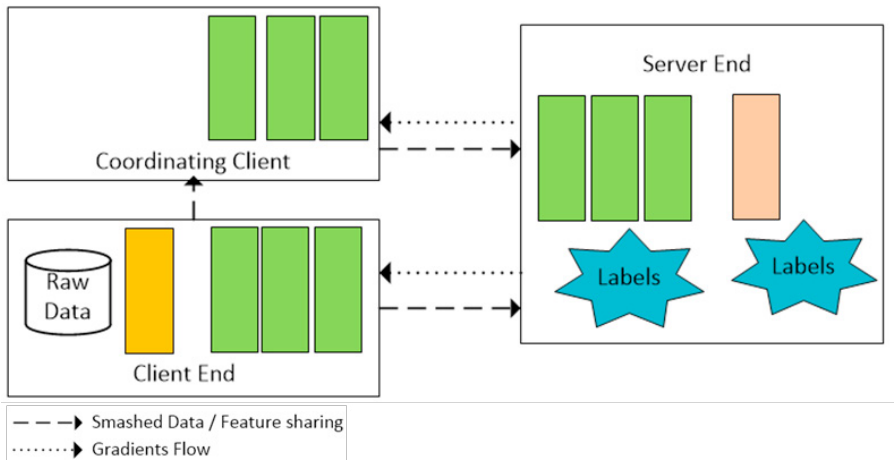


Fig. 6. Extended Vanilla Split Learning

4. Research Challenges

This section gives a brief overview of the different research challenges in designing Split learning for preserving privacy of Electronic Medical Record.

4.1. Limited Research Contribution

Since, Split Learning is an emerging technique in deep learning networks, very fewer research contributions are available. This means that multiple open research challenges are there in terms of developing newer SL configurations, modify an existing configuration, wired / wireless communication mechanisms between client and server, develop new protocol for communicating gradients to the Server, efficient and secure protocol for communicating gradients, etc. Only after developing the configurations and communication models, a precise comparison of performance can

be done with other models that uses Federated learning, Transfer learning and Conventional learning. Also, for incorporating SL in healthcare, the model has to be developed in all the aspects discussed as precision is highly important.

4.2. Split Learning Attacks

Though SL claims to be more secure than conventional deep learning model, as raw data is not shared with others, still Split learning is susceptible to certain attacks. Few of the attacks are discussed. Model Reconstruction attack / Membership inference attacks [19] [20] [21] can be termed as a white-box attack where an attacker judges or knows the working model of the client and have very limited auxiliary knowledge about the input and output data. Still, the attacker tries and succeeds in reconstructing the model and there are possibilities for finding out the raw data. Though cent percent accurate raw data can't be produced, still there is a possibility of up to 75.1% data accuracy.

Backdoor attacks [22] are termed as a type of data poisoning attack and model poisoning attacks, where few unwanted data are injected to the training model. Though the model is trained with a trigger pattern, the output still shows a normal output. These attacks compromise the training data available with the clients, either by manipulating the existing data or by poisoning the data. Also, the attacks can happen over the labels that defines the categories of the data. The label is modified without the knowledge of the client and so the client keeps on training the model based on the poisoned training data. In Model poisoning attack, the learning rule or gradients are altered and manipulated. Gradients are the one that are communicated to the server. So, making changes in the gradient value will impact the result highly and there will be a chance of the entire system to go meaningless.

There are also certain attacks related with communication. In SL, the forward and backward propagation takes place by communication between clients and server. Man- in – the – Middle (MITM) attack can be injected in the communication that takes place between the client and server. MITM attack happens by either interfering the real network or by manipulating a malicious network. Even when the gradients are encrypted, MITM attack can steal the information that is communicated. Because of all these, MITM attacks are hard to detect.

4.3. Communication overheads

When split learning is deployed in healthcare, the client machines are supposed to deal with huge volume of data as there can be images, videos and large datasets. In SL, the gradients and weights are communicated back and forth with the client and the server respectively. When multiple clients attempt to communicate with the server at the same time, communication bottleneck occurs. Also, this back propagation may occur multiple times depending on the testing accuracy. So, when multiple data are communicated multiple times, a proper secured communication protocol need to be used to handle the bottleneck and overhead.

5. Conclusion

In the modern era of AI, split learning can be used in places where there are high resource constraints. Split learning can be efficiently used for training the data that is distributed across multiple client machines. This paper reviewed the process of applying split learning for healthcare dataset that are stored in a distributed way and explained the various configurations that can be used to train the data model effectively. Though, deploying split learning has its own advantages, the paper also discusses the practical challenges in using split learning for electronic medical record. This work can be taken as a foundation for focusing the specific areas of research in split learning and improvised algorithms can be devised out in future.

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